

Assignment 2

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3.

$$(c) Q(f) = \frac{3}{2}f\left(\frac{1}{3}\right) - 2f\left(\frac{1}{2}\right) + \frac{3}{2}f\left(\frac{2}{3}\right)$$

$$f(x) = 1 \quad Q(f) = \frac{3}{2} - 2 + \frac{3}{2} = 1$$

$$\int_0^1 dx = 1$$

$$f(x) = x \quad Q(f) = \left(\frac{3}{2}\right)\left(\frac{1}{3}\right) - 2\left(\frac{1}{2}\right) + \frac{3}{2}\left(\frac{2}{3}\right) = \frac{1}{2}$$

$$\int_0^1 x dx = \frac{1}{2}$$

$$f(x) = x^2 \quad Q(f) = \left(\frac{3}{2}\right)\left(\frac{1}{9}\right) - 2\left(\frac{1}{4}\right) + \frac{3}{2}\left(\frac{4}{9}\right) = \frac{1}{3}$$

$$\int_0^1 x^2 dx = \frac{1}{3}$$

$$f(x) = x^3 \quad Q(f) = \left(\frac{3}{2}\right)\left(\frac{1}{27}\right) - 2\left(\frac{1}{8}\right) + \frac{3}{2}\left(\frac{8}{27}\right) = \frac{1}{4}$$

$$\int_0^1 x^3 dx = \frac{1}{4}$$

$$f(x) = x^4 \quad Q(f) = \frac{3}{2}\left(\frac{1}{64}\right) - 2\left(\frac{1}{16}\right) + \frac{3}{2}\left(\frac{16}{64}\right) \neq \frac{1}{5}$$

$$\int_0^1 x^4 dx = \frac{1}{5}$$

Precision $n = 4$

4.

Let π_{n+1} be the nodal polynomial associated with the quadrature points:

$$\pi_{n+1}(x) = (x - x_0)(x - x_1) \dots (x - x_n)$$

$$\text{Let } f(x) = (\pi_{n+1}(x))^2 \in P_{2n+2},$$

note that $f(x) \geq 0$ for all x , so $\int_a^b f(x) dx > 0$

but $f(x_i) = 0$ for all i so

$$I_n(f) = \sum_{k=0}^n w_k f(x_k) = 0$$

so I_n is not exact for this polynomial of degree $2n+2$
therefore doesn't have precision $2n+2$

6. $-u''(x) + u(x) = 3x$ on $(-1, 2)$ with $u(-1) = u(2) = 0$

The weak/variational formulation:

Find $u \in H_0^1(a, b)$ such that:

$$A(u, v) = L(v) \quad \text{for all } v \in H_0^1(a, b)$$

where $A(u, v) := (u', v') + (ru, v)$ and inner product $(u, v) := \int_a^b u(x)v(x)dx$
 $L(v) = (f, v)$

$$\int_a^b u'(x)v'(x)dx + \int_a^b r(x)u(x)v(x)dx = \int_a^b f(x)v(x)dx$$

Here, $b=2$, $a=-1$, $r(x)=1$, $f(x)=3x$:

Find $u \in H_0^1(-1, 2)$ such that

$$\int_{-1}^2 u'(x)v'(x)dx + \int_{-1}^2 u(x)v(x)dx = \int_{-1}^2 3xv(x)dx \quad \text{for all } v \in H_0^1(-1, 2)$$

7. $-u''(x) + u(x) = 3x$ on $(-1, 2)$ with $u(-1) = u(2) = 0$

mesh: $\{x_0, x_1, x_2, x_3\} = \{-1, 0, 1, 2\}$

The FEM is:

Find $u_h(x) \in S$ such that

$$a(u_h, v_h) := (u_h', v_h') + (u_h, v_h) = 3(x, v_h) \text{ for all } v_h(x) \in S$$

$h = \frac{2 - (-1)}{3} = 1$, so let

$$\varphi_1(x) = \begin{cases} x+1 & -1 < x < 0 \\ 1-x & 0 < x < 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\varphi_2(x) = \begin{cases} x & 0 < x < 1 \\ 2-x & 1 < x < 2 \\ 0 & \text{otherwise} \end{cases}$$

and $u_h(x) = \lambda_1 \varphi_1(x) + \lambda_2 \varphi_2(x)$

$$(\lambda_1 \varphi_1' + \lambda_2 \varphi_2', \varphi_1') + (\lambda_1 \varphi_1 + \lambda_2 \varphi_2, \varphi_1) = 3(x, \varphi_1)$$

$$(\lambda_1 \varphi_1' + \lambda_2 \varphi_2', \varphi_2') + (\lambda_1 \varphi_1 + \lambda_2 \varphi_2, \varphi_2) = 3(x, \varphi_2)$$

$$\textcircled{1} \lambda_1 \left(\int_{-1}^2 \varphi_1' \varphi_1' dx + \int_{-1}^2 \varphi_1 \varphi_1 dx \right) + \lambda_2 \left(\int_{-1}^2 \varphi_2' \varphi_1' dx + \int_{-1}^2 \varphi_2 \varphi_1 dx \right) = 3 \int_{-1}^2 x \varphi_1 dx$$

$$\textcircled{2} \lambda_1 \left(\int_{-1}^2 \varphi_1' \varphi_2' dx + \int_{-1}^2 \varphi_1 \varphi_2 dx \right) + \lambda_2 \left(\int_{-1}^2 \varphi_2' \varphi_2' dx + \int_{-1}^2 \varphi_2 \varphi_2 dx \right) = 3 \int_{-1}^2 x \varphi_2 dx$$

$$\textcircled{1} \int_{-1}^2 \varphi_1' \varphi_1' dx = \int_{-1}^0 (1)^2 dx + \int_0^1 (-1)^2 dx = \left. x \right|_{-1}^0 + \left. x \right|_0^1 = -(-1) + 1 = 2$$

$$\int_{-1}^2 \varphi_1 \varphi_1 dx = \int_{-1}^0 (x+1)^2 dx + \int_0^1 (1-x)^2 dx = \left. \frac{x^3}{3} + x^2 + x \right|_{-1}^0 + \left. x - x^2 + \frac{x^3}{3} \right|_0^1 = \frac{2}{3}$$

Coefficient of λ_1 is $\frac{8}{3}$

$$\int_{-1}^2 \psi_2' \psi_1' dx = \int_0^1 (1)(-1) dx = -x \Big|_0^1 = -1$$

$$\int_{-1}^2 \psi_2 \psi_1 dx = \int_0^1 (x)(1-x) dx = \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$$

Coefficient of λ_2 is $-1 + 1/6 = -5/6$

$$\begin{aligned} 3 \int_{-1}^2 x \psi_1 dx &= 3 \left(\int_{-1}^0 x(x+1) dx + \int_0^1 x(1-x) dx \right) = 3 \left(\frac{x^3}{3} + \frac{x^2}{2} \Big|_{-1}^0 + \frac{x^2}{2} - \frac{x^3}{3} \Big|_0^1 \right) \\ &= 3 \left(-\left(\frac{1}{3} + \frac{1}{2}\right) + \frac{1}{2} - \frac{1}{3} \right) \\ &= 0 \end{aligned}$$

$$\textcircled{1} 16\lambda_1 - 5\lambda_2 = 0$$

②

$$\int_{-1}^2 \psi_1' \psi_2' dx = -1$$

$$\int_{-1}^2 \psi_1 \psi_2 dx = \frac{1}{6}$$

Coefficient of λ_1 is $-1 + 1/6 = -5/6$

$$\int_{-1}^2 \psi_2' \psi_2' dx = \int_0^1 (1)^2 dx + \int_1^2 (-1)^2 dx = x \Big|_0^1 + x \Big|_1^2 = 1 + (2-1) = 2$$

$$\int_{-1}^2 \psi_2 \psi_2 dx = \int_0^1 x^2 dx + \int_1^2 (2-x)^2 dx = \frac{x^3}{3} \Big|_0^1 + \frac{4x - 2x^2 + \frac{x^3}{3}}{1} \Big|_1^2 = \frac{1}{3} + \left(8 - 8 + \frac{8}{3} - \left(4 - 2 + \frac{1}{3} \right) \right) = \frac{2}{3}$$

Coefficient of λ_2 is $2 + 2/3 = 8/3$

$$\begin{aligned} 3 \int_{-1}^2 x \psi_2 dx &= 3 \left(\int_0^1 x^2 dx + \int_1^2 x(2-x) dx \right) = 3 \left(\frac{x^3}{3} \Big|_0^1 + x^2 - \frac{x^3}{3} \Big|_1^2 \right) = 3 \left(\frac{1}{3} + \left(4 - \frac{8}{3} - \left(1 - \frac{1}{3} \right) \right) \right) \\ &= 3(1) \\ &= 3 \end{aligned}$$

$$\textcircled{2} -5\lambda_1 + 16\lambda_2 = 18$$

$$16\lambda_1 - 5\lambda_2 = 0$$

$$-5\lambda_1 + 16\lambda_2 = 18$$

$$\lambda_1 = \frac{5}{16} \lambda_2$$

$$-5\left(\frac{5}{16}\lambda_2\right) + 16\lambda_2 = 18$$

$$-25\lambda_2 + 256\lambda_2 = 288$$

$$231\lambda_2 = 288$$

$$\lambda_2 = \frac{96}{77}$$

$$\lambda_1 = \frac{5}{16} \left(\frac{96}{77}\right)$$

$$= \frac{30}{77}$$

$$u_n = \lambda_1 \varphi_1(x) + \lambda_2 \varphi_2(x) = \frac{30}{77} \varphi_1(x) + \frac{96}{77} \varphi_2(x)$$